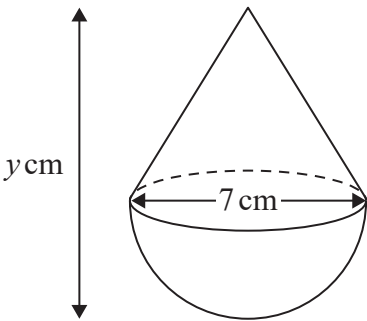
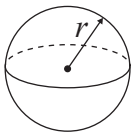


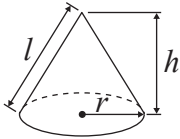
17 A solid cone is joined to a solid hemisphere to make the solid **T** shown below.



Volume of sphere = $\frac{4}{3}\pi r^3$



Volume of cone = $\frac{1}{3}\pi r^2 h$



The diameter of the base of the cone is 7 cm.
The diameter of the hemisphere is 7 cm.

The total volume of **T** is $120\pi\text{cm}^3$
The total height of **T** is $y\text{cm}$.

- (a) Calculate the value of y .
Give your answer correct to 3 significant figures.

$y =$

(4)

The diameter of the base of the cone and the diameter of the hemisphere are both increased by the same amount.
Assuming the total volume of **T** does not change,

- (b) explain the effect this would have on your answer to part (a).